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The Motor Cortex and Corticospinal Tract of the Armadillo (*Dasypus novemcinctus*)

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INTRODUCTION

To our knowledge, cortical stimulation studies have not been published on any representative of the edentate line of mammals. For this reason such a study was undertaken on the armadillo, *Dasypus novemcinctus*. The preliminary stimulation results were surprising in the light of certain anatomical facets of the armadillo cortex and corticospinal tract previously reported by Fisher, Harting, Martin and Stuber (1969). Consequently, it was decided to combine an investigation on the stimulation of the armadillo cortex with a re-evaluation of the cytoarchitecture of the motor cortex and the anatomy of the corticospinal tract.

MATERIALS AND METHODS

Six adult armadillos ranging in weight from 3–5 kg were utilized for the stimulation portion of this study. The animals were anesthetized for surgery by administering either sodium pentobarbitol (30 mg/kg of body weight) or ether. They were maintained in a deep plane of anesthesia during the surgery, but were allowed to become very light prior to stimulation.

Each animal was placed in a Stoelting stereotaxic apparatus with the limbs dependent. The skin caudal to the hard epidermal plate over the skull was incised in order to insert a pair of large scissors. Heavy scissors were used to cut through the epidermal plate. The thin underlying connective tissue was subsequently incised to expose the calvarium. In all cases the bone caudal and rostral to the supraorbital sulcus (Smith 1899, see Fig. 1) was removed. Access to the presupraorbital cortex was gained by employing rongeurs to remove the bone encasing the frontal air sinus. After the bone

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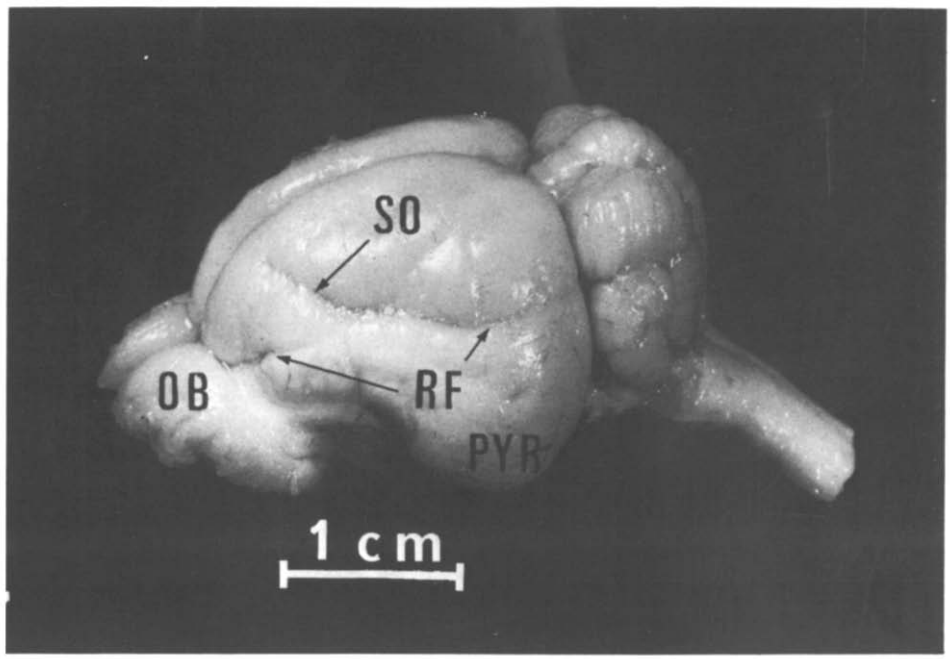


Fig. 1. A photograph of the lateral aspect of the gross brain of the armadillo. The supraorbital sulcus (SO), the rhinal fissure (RF), the olfactory bulb (OB) and the pyriform cortex (PYR) are labeled for reference.

was removed, the dura mater and arachnoid were incised and the cortex was kept moist with warm saline. In three of the experiments the cortex was protected by an oil bath.

The cortex was stimulated using either concentric or conventional bipolar electrodes. The conventional bipolar electrodes consisted of two fine gauge stainless steel wires that were 1 mm apart at the stimulating tip. A Grass S-8 stimulator connected to two Grass 4678 stimulus isolation units was used to deliver symmetrical biphasic stimuli at a rate of 62 per sec for a duration of 2 sec. In all cases a 2 min rest period was allowed between successively applied trains of stimuli and the current was controlled so as not to exceed 1.0 mA. Each point was stimulated with varying amounts of current and at every point an attempt was made to find the minimum current which would elicit a response.

Three assistants were used in each stimulation experiment to keep a running record of the results. By recording each stimulated point and the movements obtained at that point a map of the electrically-excitable motor cortex was plotted. All observations regarding the types of movements elicited and their respective thresholds were recorded.

The histology of the cortex rostral and caudal to the supraorbital sulcus (Fig. 1) was studied by employing the thionin method on 4 normal brains cut either in the frontal or the sagittal plane. Thirteen armadillos were used for the degeneration portion of the present study. The surgical approach was the same as that described previously. Lesions were placed either by gentle suction or by cautery in the cortex just rostral to the supraorbital sulcus (Fig. 1, 5 animals), in areas just caudal to the same sulcus

(6 animals), in the temporal area of the neocortex (1 animal) and in the caudal pole of the neocortex (1 animal). The animals were allowed to survive for 7–14 days before being sacrificed by perfusion. Selected segments from the removed spinal cords were processed either by the Nauta-Gygax (1954) or Fink-Heimer method (1967) for demonstrating degenerating axons. In order to describe fiber terminations in terms of Rexed's (1952) laminae, adjacent sections of each segment were stained by the thionin method. The impregnated spinal cord sections were enlarged by use of a Bausch and Lomb trisimplex projector and drawn on Bristol board. The location of degenerating fibers within each section was plotted and subsequently transposed to the drawings of the sections. Degenerating fibers were judged to be terminal when they left the main pathway and coursed in a random fashion, suggesting that they ended within the neuropile. In each case the forebrains were processed either by one of the silver methods or by the thionin technique in order to observe the full extent of the ablation.

RESULTS

The results obtained in this study will be described under three divisions: (a) results of stimulation studies, (b) histology of the cortex adjacent to the supraorbital sulcus and (c) results of degeneration studies.

Results of stimulation studies

Fig. 2 is a composite of the observations obtained in all 6 animals. The results were consistent in location, but every movement was not necessarily elicited in all animals.

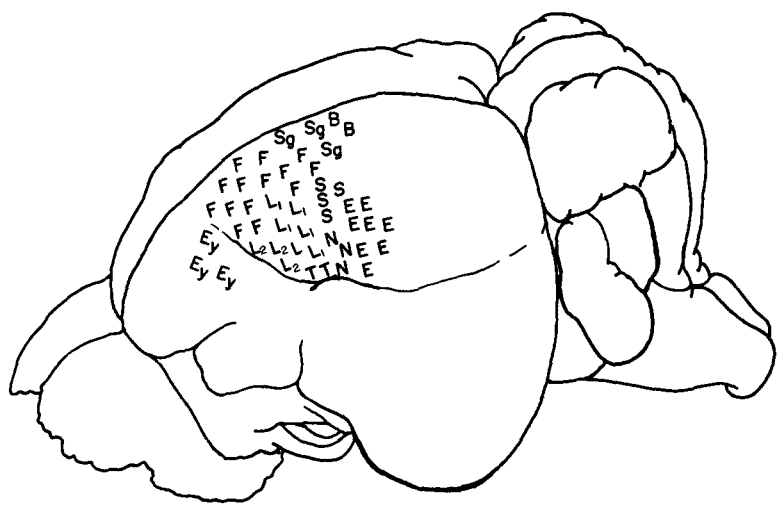


Fig. 2. Drawing of a lateral view of the gross brain. The results of cortical stimulation are indicated as follows: B, contraction of paravertebral musculature; E, movements of pinna of ear; Ey, eye movements; F, forelimb movements; L_1, movements of upper lip; L_2, movement of lower lip; N, movements of nose; S, scalp movements; Sg, movements of scapula; T, movements of tongue.

Although the motor eye fields were located in the presupraorbital cortex (Ey in Fig. 2), the supraorbital sulcus formed the rostral boundary for the majority of the electrically-excitable somatic motor cortex. The motor eye fields were not studied in detail. Stimulation immediately dorsal to the rhinal fissure and directly caudal to the ventral part of the supraorbital sulcus produced movements of the lower lip (L_2 in Fig. 2), the tongue (T in Fig. 2) and the nose (N in Fig. 2). Stimulation just dorsal and medial to the lip area resulted in movements of the rostral extremity (F in Fig. 2). These movements were limited to extension of the "wrist" and abduction of the digits. Immediately caudal to the nose area, there was a cortical region which when stimulated gave rise to movements of the pinna of the ear (E in Fig. 2). The ear area overlapped rostrally and medially with the cortical scalp representation (S in Fig. 2), so that ear movements often masked the scalp movements. Movements of the scapula (mainly retraction) were elicited from the dorsomedial extreme of the cortex caudal to the forelimb region (Sg in Fig. 2), whereas the paravertebral muscles responded to stimulation of the cortex caudal to the area for scapula movements (B in Fig. 2). Hind-limb movements were obtained only from the medial side of the hemisphere. The hind-limb movements were inconsistent, weak and limited to fanning of the toes and flexion at the hip joint. Although not distinct, movements of the distal portions of the extremity were obtained rostral to those of the hip.

Motor responses were limited mainly to the contralateral side; although, in a few instances, either bilateral or exclusively ipsilateral responses were obtained. In either case, the response elicited corresponded to the cortical representation shown in Fig. 2. Fairly consistent bilateral responses were obtained upon stimulation of the cortical area which gave rise to movements of the external ear. However, the contralateral response was still the strongest. It should be emphasized that there were no sharp boundaries between the representation of different body parts on the armadillo motor cortex. In all cases, overlap was present.

The cortex related to the lips, nose and tongue had the lowest threshold of all the regions stimulated (0.2 mA). The forelimb and external ear areas required a current ranging from 0.25–0.3 mA to elicit a response, whereas the cortex related to the caudal extremity had the highest threshold and required a current of 0.5–0.7 mA before movements could be evoked. The eye fields responded only to current in excess of 0.7 mA. Although the entire neocortex was stimulated in the present study, no evidence was found for a supplementary motor area.

Histology of the cortex adjacent to the supraorbital sulcus

Grossly the supraorbital sulcus appeared to be a rostral and dorsal extension of the rhinal fissure (Fig. 1). However, thionin-stained frontal sections revealed that the cortex rostral and ventral to the supraorbital sulcus (presupraorbital area) was not comparable to pyriform cortex. The transition from presupraorbital cortex to pyriform cortex could be readily appreciated in stained sections even though frequently there was no distinct sulcus between the two regions. The presupraorbital cortex was characterized by a wide internal granular layer (250 μm) and the relative lack of deeply-staining pyramidal neurons (A in Fig. 3). The characteristic round, relatively light- staining neurons of the internal granular layer measured up to 27 μm

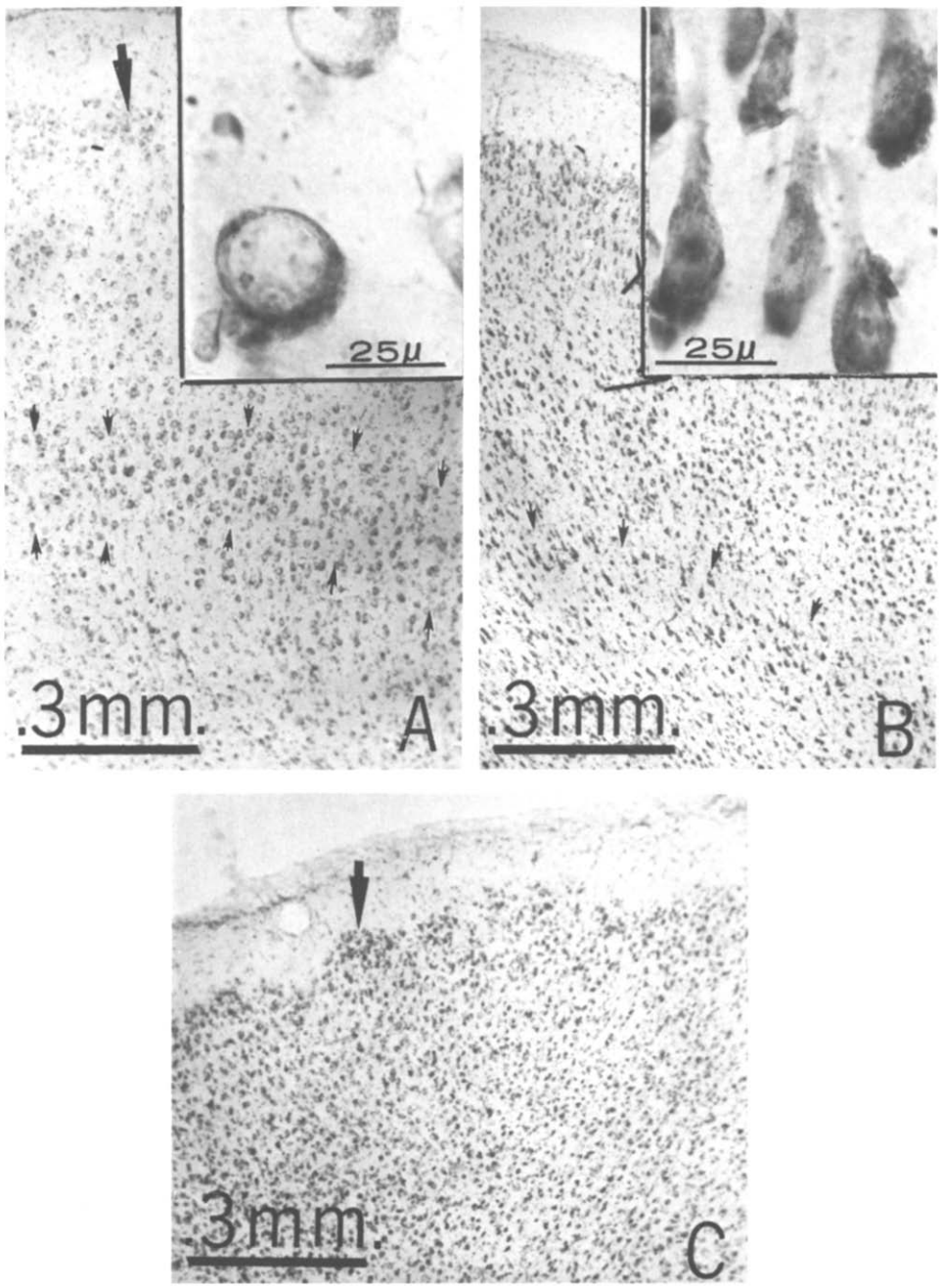


Fig. 3. Low-power photomicrographs of the presupraorbital cortex (*A*), the forelimb region of the postsupraorbital cortex (*B*) and the external layers of the face area of the postsupraorbital cortex (*C*). The insert in *A* is a high power photomicrograph of a typical cell from the internal granular layer. The insert in *B* is a high-power view of several pyramidal neurons located in the forelimb region of the motor cortex. The large arrow in *A* points to a cluster of cells in the external granular layer, whereas the smaller arrows in *B* point to the largest deep-staining pyramidal cells in the forelimb area of the postsupraorbital cortex. The large arrow in *C* points to a cluster of cells in the outer granular layer. Thionin method.

in diameter (inset in A of Fig. 3). The external granular layer was not as distinct under low power as that of the pyriform cortex because the neurons were less densely packed. In some sections, the neurons of the external granular layer of the presupraorbital cortex were situated in loose aggregates separated in part from one another by fiber bundles (A in Fig. 3). The cortex deep to the supraorbital sulcus was very thin (0.5 mm thick as compared to 1.5 mm for presupraorbital cortex and 2 mm for postsupraorbital cortex). In contrast to the presupraorbital cortex, the postsupraorbital cortex was relatively agranular (particularly dorsally) and was characterized by the presence of deeply-staining pyramidal cells (B in Fig. 3). The scattered pyramidal cells in the face area (ventral postsupraorbital cortex) were approximately 25 μm in their widest dimension, whereas those in the fore-limb area (dorsal postsupraorbital cortex) were more numerous and generally somewhat larger (around 35 μm, inset in B of Fig. 3). The largest pyramidal cells of the limb area of the motor cortex were approximately 50 μm at their widest dimension and were located in the deepest portion of layer V. The pyramidal cells of the fore-limb area were more tapering than those of the face region and their apical dendrites were apparent for a greater distance. A large number of pyramidal cells was also present on the medial aspect of the neocortex (hind-limb area). They were generally comparable in size and number to those of the fore-limb region.

The external granular layer of the face region of the postsupraorbital cortex was similar in appearance to that of the presupraorbital area *i.e.*, the neurons were located in aggregates at least partially separated from one another by fibers (C in Fig. 3). This arrangement of neurons was not present in either the fore-limb or hind-limb areas of the neocortex.

Results of degeneration studies

Lesions restricted to regions either rostral to the supraorbital sulcus or at the caudal or ventrocaudal extremes of the neocortex did not result in discernible fiber degeneration within the spinal cord. However, in those animals with lesions which included the fore-limb area of the electrically excitable motor cortex, degenerating fibers were observed in the spinal cord as far caudally as rostral thoracic levels (Figs. 4, 5). Such fibers were very small and thinly myelinated (if at all) and coursed either in the most dorsal portion of the contralateral ventral funiculus (intracommissural bundle seen in Figs. 4, 5A, B) or within the contralateral lateral funiculus (Figs. 4, 5A, C). They did not course in compact bundles, but were scattered among much larger, myelinated fibers (Fig. 5B, C). A few degenerating fibers were present in comparable areas on the side of the ablation. At cervical levels, the most extensive terminal degeneration was located within the medial portions of Rexed's laminae IV through VI (Figs. 4, 6). Some fiber degeneration was present in the medial part of lamina III and the lateral part of laminae V and VI (Fig. 4). Although a few degenerating fibers were present in laminae VII and VIII at rostral cervical levels, they were not apparent in the cervical enlargement. The location of the lesion within the fore-limb area of the motor cortex did not appear to make any difference in the pattern of fiber degeneration within the spinal cord gray matter. Cortical lesions, regardless of their size or location did not produce overt ambulatory deficits in the armadillo.

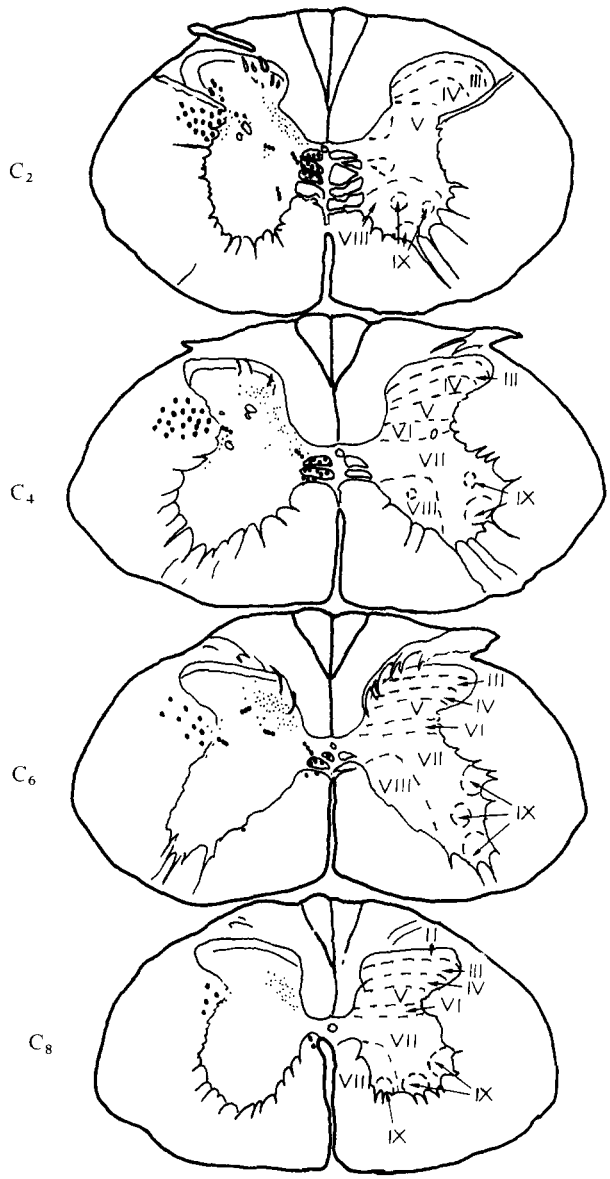


Fig. 4. A series of drawings of representative sections of the cervical spinal cord of the armadillo (C_2, C_4, C_6, C_8). Rexed's laminae are outlined and numbered on the right hand side. Degenerating fibers of passage (large dots) and those judged to be terminal (small dots) are present on the left-hand side of each section.

DISCUSSION

Cortical stimulation studies on relatively small brains such as that of the armadillo must be interpreted with care. It is difficult to establish accurately the size of an area of representation because the stimulating current may activate areas removed some distance from the site of the stimulation. Facilitation is also a problem on a small cortical surface since one cannot stimulate sequential points which are well separated

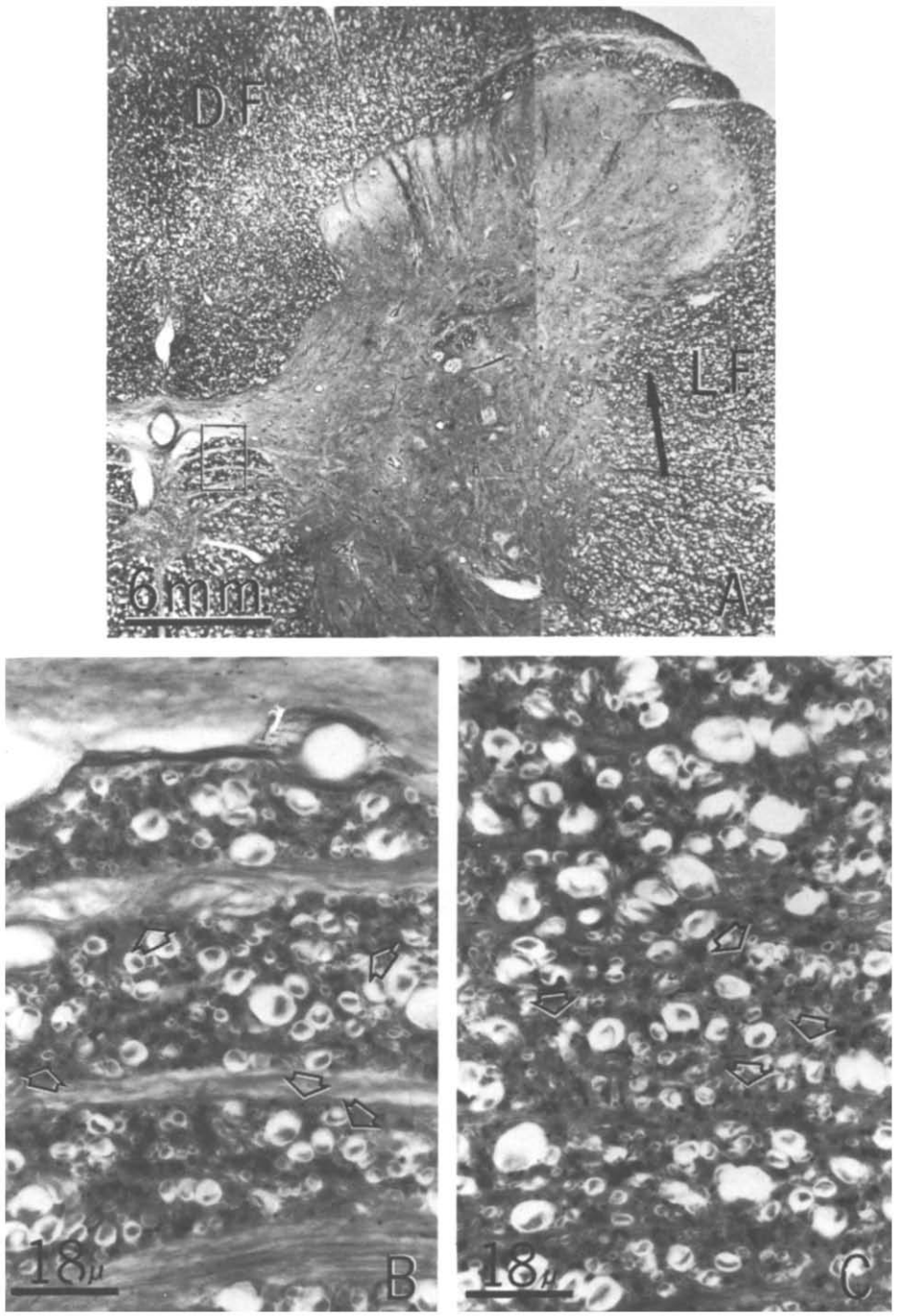


Fig. 5. A low-power photomicrographic montage from the first cervical level (*A*) of an animal with a large postsupraorbital lesion. The dorsal funiculus (D.F.) and the lateral funiculus (L.F.) are indicated for reference. The rectangle at the lower left outlines the area of the ventral funiculus shown in high power at the lower left (*B*), whereas the arrow at the right points to the region of the lateral funiculus shown in higher power in the photomicrograph on the lower right (*C*). The small open block arrows in *B* and *C* point to the small, impregnated corticospinal fibers cut in cross-section. Note that they are scattered among larger normal myelinated fibers. Fink–Heimer technique.

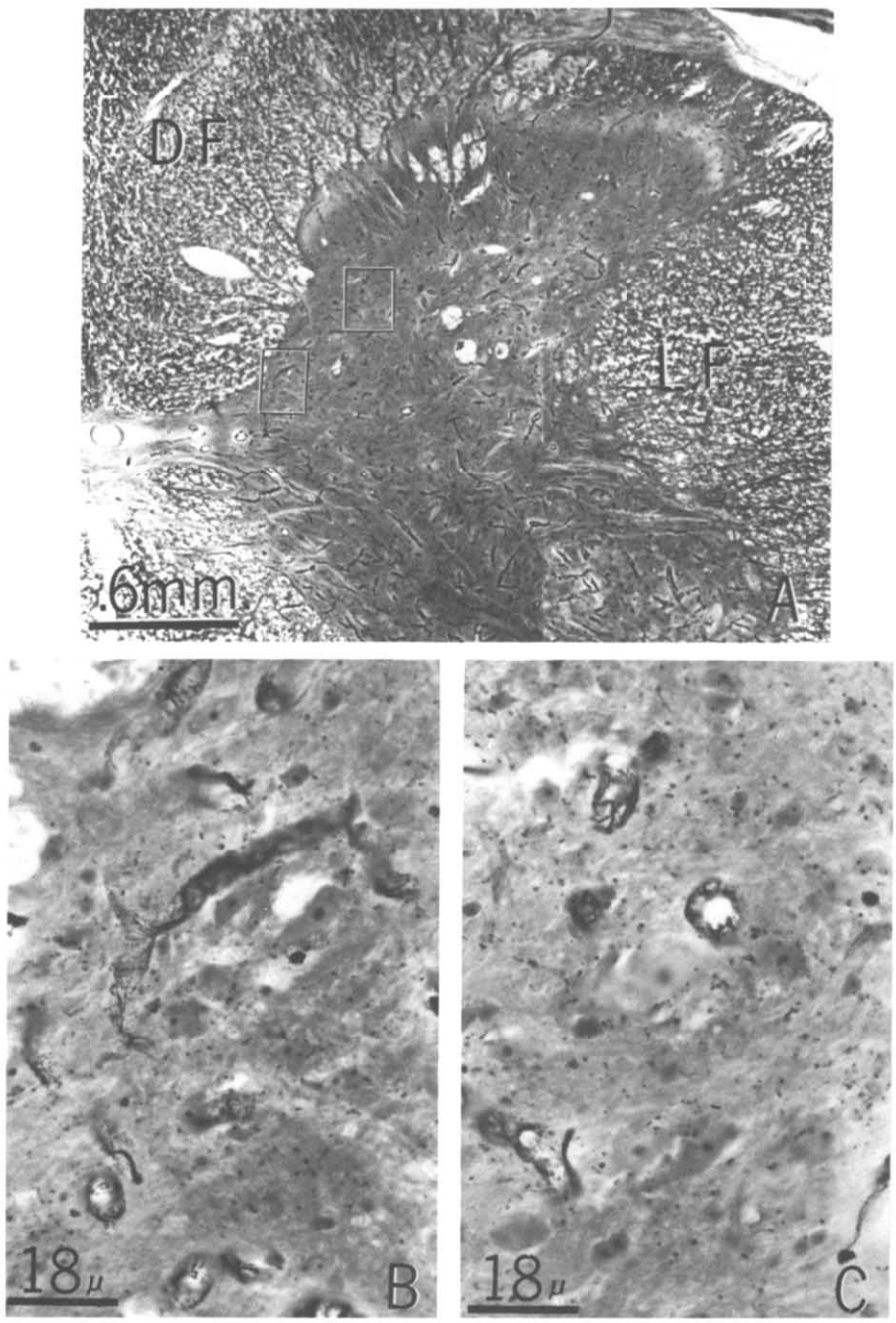


Fig. 6. A low-power photomicrographic montage of C_6 (*A*) from an animal with a large postsupraorbital lesion. The dorsal funiculus (D.F.) and lateral funiculus (L.F.) are indicated for reference. The lower left rectangle outlines the area of lamina VI shown in higher power in *B*, whereas the upper right rectangle indicates the portion of lamina V (mainly) shown in *C*. Degenerating fibers are present in both of the lower photomicrographs (*B*, *C*). Fink–Heimer technique.

from one another. For these reasons, diminutive areas of representation might not be isolated by stimulation in small brains and this might help to account for the negative finding relating to a supplementary motor area in the armadillo.

In spite of the limitations of this kind of cortical stimulation study, a scheme of body representation was apparent in the electrically-excitable motor cortex of the armadillo. The pattern of somatotopic representation corresponded to the general mammalian pattern described for the cat, the rabbit, the rat and the monkey by Woolsey (1958). The degree of cortical representation of a body part in either motor or sensory cortex of an animal often reflects the functional specialization of that part. This is emphasized particularly in the digital representation on the motor cortex of the raccoon (Hardin, Arumugasamy and Jameson 1968). Examination of the motor map of the armadillo suggests that the representation for the lips, tongue and snout is disproportionately large when compared to the rest of the body. As might be expected, these areas are employed extensively by the armadillo to obtain food. The relatively small limb representation is not surprising since the digits are not utilized individually and, although the limbs are powerful, they are used mainly for digging and clawing. In spite of the large face representation on the motor cortex of the armadillo, the motor cortex does not project directly to cranial nerve motor nuclei (Harting and Martin 1970).

The hind-limb cortical representation is interesting in the light of the lack of corticospinal fibers which extend to the lumbosacral spinal cord. The hind-limb movements obtained by stimulating the medial side of the hemisphere must be a result of cortical activation of subcortical centers either in the brain stem (red nucleus and/or the reticular formation) or within the spinal cord which in turn relay to the lumbosacral spinal cord. This suggestion is reinforced by the somewhat higher threshold to stimulation for the hind-limb area as compared to the fore-limb region. Such is apparently also true for the opossum (compare Lende 1963 with Martin and Fisher 1968).

The histology of the armadillo motor cortex fits the generally accepted mammalian pattern. As might be expected on the basis of the stimulation results, the cortex caudal to the supraorbital sulcus contains a large number of deeply-staining pyramidal neurons, particularly in the areas of limb representation, whereas the presupraorbital cortex contains relatively few such neurons. It would appear that these cortical areas were reversed in the description by Fisher *et al.* (1969).

In the present study, the results of the degeneration study compare favorably with those obtained from cortical stimulation. However, the lack of body movements from stimulating presupraorbital cortex and the absence of degeneration within the corticospinal tract after presupraorbital lesions are in disagreement with the results of a previous study on the armadillo corticospinal tract by Fisher *et al.* (1969). These authors claimed that presupraorbital lesions resulted in fiber degeneration within the corticospinal tract which was distributed to more lateral areas of the spinal cord gray than that observed after postsupraorbital lesions. Re-examination of the material from which the study by Fisher *et al.* (1969) was taken revealed that the spinal cord tissue taken from the animals with presupraorbital lesions contained a considerable amount of silver precipitate thereby increasing the possibility of misreading the slides.

The distribution of corticospinal fibers to the medial part of the dorsal horn is

similar to the distribution of such fibers in the opossum (Bautista and Matzke 1965; Martin and Fisher 1968) and the tree shrew (Shriver and Noback 1967; Jane, Campbell and Yashon 1969). The area of termination of such fibers suggests that the corticospinal system may have a particularly significant role as a cortical feedback system on sensory information in these forms.

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SUMMARY

The neocortex of the armadillo, *Dasypus novemcinctus*, was explored by electrical stimulation and the motor area determined to be caudal to the supraorbital sulcus. Movements of the tongue, lips, pinna, scalp, shoulder, back, fore-limb and, to some extent, hind-limb could be elicited. The motor area for the lips and tongue was particularly large and it had the lowest threshold to stimulation. The motor cortex was characterized by deeply-staining pyramidal cells and the lack of a well-developed internal granular layer. The pyramidal cells were largest and most numerous in the limb area.

Ablation of the motor cortex or part of it resulted in fiber degeneration within the contralateral ventral and lateral funiculi of the spinal cord. Degenerating fibers distributed extensively to the medial parts of laminae IV through VI and, to a much less extent, to the medial part of lamina III and the lateral parts of laminae V and VI. Degenerating fibers were not observed in the spinal cord after presupraorbital lesions or after ablations of the caudal or ventrocaudal neocortex.

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